

Veille informatique quantique - 2026-09-25

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Quantum computational advantage in random-circuit sampling on IBM superconducting quantum computers

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.28657v1 Announce Type: new Abstract: We report forward random-circuit sampling (RCS) on the 120-qubit Nighthawk r2 superconducting processor (*ibm_phoenix*) with square-lattice connectivity, using 61 qubits, native CZ gates, and the standard cloud execution stack with no benchmark-specific calibration. Two independent fidelity estimators---mirror benchmarking and three- and four-patch cross-entropy benchmarking (XEB)---agree with each other at every measured depth, the mirror

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit, quantum processor.

2. Nonorthogonal variational quantum simulation for quantum chemistry

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.29337v1 Announce Type: new Abstract: Dynamical simulation of quantum many-body systems is a central task in quantum chemistry and requires efficient wavefunction representations. Although tensor-network and neural-network quantum states have achieved considerable success in ground-state calculations, entanglement growth hinders their application to quantum dynamics. Quantum computing may offer a route to quantum advantage, but algorithms such as Trotterization generally require fault

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3. Design Principles for Ultra-High-Rate Quantum Codes

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.30069v1 Announce Type: new Abstract: Reducing the qubit overhead of quantum error correction is a central challenge for scalable fault-tolerant quantum computing. Recent ultra-high-rate quantum codes offer a promising route toward this goal, with some constructions requiring as few as two physical data qubits per logical qubit. However, systematic principles for navigating the tradeoffs among encoding rate, distance, check weight, and blocklength remain lacking. Here, we develop and

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Mots-clés détectés : quantum computing, qubit, quantum error correction.

4. Fundamental Physics at the Frontier of Noisy Quantum Computation

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.28825v1 Announce Type: new Abstract: Quantum computing offers a new, orthogonal direction for investigating fundamental physics, extending beyond classical numerical methods and conventional observables. Realizing this potential requires directly confronting the noise limiting currently available quantum computers. Progress rests on advancing algorithms, interpreting their results, and managing their errors together. This thesis presents several advancements in the use of quantum sim

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Mots-clés détectés : quantum computing, quantum computer.

5. Reconfigurable bus-based quantum router for modular superconducting processors

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.28881v1 Announce Type: new Abstract: Scaling superconducting quantum processors requires interconnects that provide both non-local connectivity and parallel entangling operations. Nearest-neighbour couplings require distant interactions to be routed through SWAP networks, increasing the native two-qubit-gate count and potentially extending the circuit critical path. Here we introduce a bus-based reconfigurable quantum router for modular superconducting processors. Flux-tunable SQUID

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Mots-clés détectés : qubit, quantum processor.

6. On the numerical limitations of dual Koopman von Neumann embeddings for solving conservative nonlinear ordinary differential

equations on quantum computers

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.28999v1 Announce Type: new Abstract: The simulation of nonlinear ordinary differential equations on quantum computers is inherently challenging, as quantum gates are linear operators on qubit states. In this paper, we put forth a Koopman-von Neumann (KvN) operator based algorithm for solving nonlinear ordinary differential equations on a quantum computer which overcomes the innate limitations of quantum operations. In this approach, a Liouville probability density is embedded into a

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Mots-clés détectés : quantum computer, qubit.

7. Efficient Synthesis of Multi-Controlled Toffoli Gates with Ternary Clifford+P₉ Gates

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.29884v1 Announce Type: new Abstract: Temporary occupation of qutrit levels can reduce the width and depth required for binary circuit logic in quantum computing. We introduce an efficient intermediate-qutrit decomposition of multi-controlled Toffoli gates with binary-subspace inputs and outputs. For balanced control widths $n=2^h-1$, the proposed decomposition is evaluated hierarchically through a tree, allowing independent subtree computations to proceed in parallel and reducing the

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Mots-clés détectés : quantum computing, quantum algorithm.

8. Loan Portfolio Optimization with Variational Quantum Algorithms

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.30195v1 Announce Type: new Abstract: Loan portfolio optimization (LPO) seeks portfolios that minimize credit risk while satisfying practical selection constraints. Accounting for portfolio-level risk requires modeling not only expected losses but also loss variability and borrower-default correlations, leading to a large-scale combinatorial optimization problem whose complexity grows rapidly with portfolio size. In this work, we formulate LPO as a Quadratic Constrained Binary Optimiz

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Mots-clés détectés : qubit, quantum algorithm.

9. A Block Decomposed QUBO Workflow for Chromosome-Y Phylogeny Reconstruction

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.29856v1 Announce Type: cross Abstract: This paper sets out a computational workflow that reconstructs the phylogeny of human Y-chromosome populations from a Variant Call Format (VCF) file of biallelic Single Nucleotide Polymorphisms (SNP). The two classical phylogenetic decisions - topology selection and root placement - are cast as Quadratic Unconstrained Binary Optimisation (QUBO) problems. The workflow combines two QUBO formulations with an Alternating Direction Method of Multipli

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Mots-clés détectés : quantum computing, qubit.

10. Non-Perturbative Topological Gadgets for Many-Body Coupling

Source : [arXiv Quantum Physics](#) - publié le 2026-09-25

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2506.05323v3 Announce Type: replace Abstract: Continuous-time quantum hardware implementations generally lack the native capability to implement high-order terms that would facilitate efficient compilation of quantum algorithms. This limitation has, in part, motivated the development of perturbative gadgets---multi-qubit constructions used to effect a desired Hamiltonian using engineered low-energy subspaces of a larger system constructed using simpler, usually two-body, primitives. In th

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Méthode

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